

Master's Programme in Electronics and Nanotechnology

Evidence-Grounded Processing and Penetration Prediction of Mie-Scattering Spray Videos

From calibration and feature extraction to baseline and uncertainty-aware
surrogate models

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Abstract

This thesis presents a computational workflow for predicting the penetration evolution of pilot diesel spray in multi-fuel marine engines and for providing geometric input to early-stage impingement-risk screening. The motivation is to support early engineering assessment of liquid diesel interaction with piston or cylinder surfaces, because such interaction may contribute to lubricant dilution, wall wetting, and poor mixture preparation.

The workflow is based on room-temperature Mie-scattering spray experiments recorded in an Optical Spray Combustion Chamber with high-speed imaging. The data consist of terabyte-scale 12-bit grayscale videos collected with a fixed camera setup for one injector under multiple operating conditions and nozzle geometries. A manual calibration step first determines the spray center and the pixel-to-length scaling. The calibrated videos are then processed by a CUDA-accelerated workflow that extracts plume-resolved temporal observables and boundary evolution from the raw image sequences.

To construct learning targets, penetration trajectories are represented with a reduced-order empirical fit. This step mitigates right-censoring caused by the finite observation window, improves temporal alignment across conditions, and produces representative penetration targets. On this basis, a multi-stage neural-network surrogate is developed: a deterministic baseline is trained on median penetration trajectories, a heteroscedastic model is trained on the full filtered fitted-trajectory set, and a teacher-guided refinement stage uses direct raw time-series supervision where reliable observations are available while retaining conservative teacher guidance in sparse regions. This staged design avoids the systematic underestimation bias that would arise from training a single model directly on right-censored observations with sparse or incomplete labels. The predicted mean and uncertainty can then be combined with simplified spray-angle assumptions and cylinder masks as a prototype route for probabilistic impingement screening.

The main outcome is an engineering-oriented surrogate workflow that connects large-scale optical spray imaging to penetration prediction and geometric risk screening. Evaluated across approximately 15,000 penetration trajectories from 9 nozzle

campaigns (roughly 400,000 labelled time steps), the refined surrogate achieves an overall RMSE of 6.9 mm (7.2% of the measured penetration range), with heteroscedastic prediction intervals calibrated to nominal coverage (68.5% at 1σ , 95.7% at 2σ). The method is intended as a front-end screening tool, not as a complete physical truth model of in-cylinder spray dynamics. Its current scope is limited to room-temperature non-reacting data and simplified collision geometry, but it provides a basis for later physically regularized and higher-dimensional extensions.

Keywords Mie scattering, spray imaging, penetration, computer vision, surrogate modeling

Preface

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1 Introduction

1.1 Research relevance

International shipping is under increasing pressure to reduce greenhouse-gas emissions while maintaining fuel flexibility and operational reliability. The 2023 IMO greenhouse-gas strategy sets both continued carbon-intensity reduction targets and an uptake target for zero- or near-zero-emission energy by 2030, which has accelerated practical interest in alternative-fuel propulsion systems [15]. Industry tracking also shows that liquefied natural gas, methanol, ammonia, and other alternative fuels have already moved from concept discussion into concrete vessel and orderbook decisions [4].

Within that transition, dual-fuel and multi-fuel marine engines depend not only on fuel availability but also on reliable ignition and stable combustion. For many of these engines, a small pilot-fuel injection provides the ignition source for the main fuel. In low-pressure gas operation, for example, WinGD X-DF engines use a small pilot-fuel quantity for ignition and can revert to diesel mode if gas supply is lost [49, 48]. Similar logic extends to methanol and ammonia concepts, where pilot fuel remains necessary because the main fuel does not readily auto-ignite under the same conditions [5, 47].

However, pilot diesel spray is not only an ignition aid. Its spatial development also matters. If liquid fuel reaches the piston bowl or cylinder wall before forming a

useful ignition kernel, wall wetting, deteriorated combustion, higher emissions, and lubricant dilution may follow [35, 40]. For injector development and early design screening, it is therefore valuable to estimate the liquid spray envelope and its potential impingement risk before moving directly to costly engine testing or high-fidelity combustion simulation.

1.2 Current engineering practice

In engineering practice, spray behavior and impingement risk are usually assessed through three broad routes: engine test campaigns, computational fluid dynamics and spray-combustion simulations, and optical spray experiments. Engine tests are closest to real operation, but they are expensive, slow, and difficult to use for isolating one spray-design factor at a time. High-fidelity simulations can provide richer internal flow information, but they require complex modeling choices, many input assumptions, and substantial computational effort [24, 35]. This cost is not only qualitative. In a reacting diesel-spray LES example cited in combustion-computing literature, one high-resolution realization with up to 22 million cells and a 54-species *n*-dodecane mechanism required about 48,000 CPU core-hours and up to 20 days of wall-clock time for only 2 ms after start of injection [7]. Because TFLOP/s depends strongly on hardware and solver efficiency, core-hours are a more reproducible cost measure; even a conservative per-core throughput conversion places this example at petaflop-order arithmetic per realization, before any condition sweep.

Optical experiments occupy a different role. They provide controlled and repeatable observations that are well suited to comparing many candidate injectors, settings, or operating points. At the same time, optical data do not automatically translate into engineering decisions. In particular, liquid-phase boundaries observed in spray imaging depend on the diagnostic method and on post-processing choices. ECN guidance explicitly notes that liquid-penetration measurements are method-dependent and that optical boundaries should be interpreted as experimentally defined quantities rather than as direct access to a hidden physical ground truth [10, 9].

1.3 Existing solutions

Diesel-spray research already offers a broad set of experimental and modeling approaches. On the experimental side, high-speed imaging and optical diagnostics are widely used to characterize penetration, spreading angle, and spray morphology. On the modeling side, empirical correlations, phenomenological formulations, and CFD-based methods are commonly used to describe spray development [23, 13, 30, 24]. These approaches have established an important foundation for injector design and combustion-system analysis.

Nevertheless, each class of solution has limitations for the present use case. Empirical and phenomenological models are attractive because they are compact and interpretable, but their assumptions are often strongest under quasi-steady conditions and become less reliable under short-pulse, strongly transient, or multiple-injection cases [24]. In addition, many classical formulations are piecewise or regime-switched

and therefore not especially smooth as learning targets, which makes them less suitable for fitting noisy short-duration pilot-spray trajectories. High-fidelity CFD is information-rich, yet it is rarely the most efficient tool for early-stage screening across many design candidates. Recent LES studies also treat computational cost as a central design constraint; for example, one ducted-fuel-injection LES study reports that a multi-slice ensemble strategy saved about 447,000 core-hours for one configuration compared with a more direct sampling approach [38]. Purely image-based measurements, in turn, describe what is observed in a specific optical setup, but they do not by themselves provide a predictive intermediate-layer model for engineering comparison.

More recently, machine-learning methods have begun to appear in spray modeling and combustion-surrogate literature as a complement to classical empirical and phenomenological models. These approaches demonstrate that learned regression can reproduce gross spray behavior without a closed-form analytical law. However, the specific combination of constraints in the present work has not been the focus of those efforts: short-pulse pilot injections whose transient regime dominates the observation window, right-censored measurement trajectories produced by a finite-view chamber, and uncertainty-aware outputs intended for downstream geometric risk screening. This gap between general data-driven spray regression and the requirements of the present problem motivates the purpose-built surrogate workflow described in this thesis.

1.4 Problem statement

The central problem addressed in this thesis is therefore not whether diesel spray has been studied, but whether a practical intermediate-layer model can be built between optical experiments and high-cost validation. More specifically, the missing element is a fast and repeatable method for predicting the time-evolving non-reacting liquid envelope of pilot diesel spray so that later impingement-risk analysis has a defensible geometric input. The target is not to replace CFD or engine testing, but to introduce a surrogate layer between optical experiments and those higher-cost validation routes.

This gap matters because short pilot injections are strongly transient. Spray angle and penetration cannot always be treated as fixed constants, and transient spreading behavior influences downstream penetration and mixing [18]. At the same time, spray-wall interaction remains physically important and not fully resolved, particularly when one moves from laboratory diagnostics toward engine-relevant consequences such as wall film formation and reliability concerns [35]. In marine engines, the situation is further complicated by geometry and scale, because large-bore conditions may differ from the more familiar automotive liquid-impingement picture [25]. What is currently lacking is not another isolated image measurement, but a practical predictor that supports early risk screening before full reacting-spray or engine-level analysis.

One promising route is to learn this predictor directly from large-scale optical spray data. The key observation is that Mie-scattering experiments already capture the relevant liquid-envelope behavior across a range of conditions; what is missing is a way to turn those observations into a fast, condition-parametric model. A data-driven surrogate trained on reduced-order fits to measured penetration trajectories

occupies exactly the gap between raw experimental measurements and physics-based simulation: it retains the empirical fidelity of the optical data while providing the generalization and computational speed needed for engineering screening. By learning the mapping from injection conditions directly to penetration trajectory parameters, the surrogate bypasses the need for a closed-form spray law and remains adaptable as the dataset grows. Uncertainty quantification further makes the outputs defensible for risk assessment, in contrast to a single deterministic prediction that conveys no information about reliability. This requirement is not only conceptual. Mie-scattering experiments exhibit inherent shot-to-shot variability, optical boundary extraction depends on threshold choices whose sensitivity varies across operating conditions, and the number of usable repeat injections per point is limited by the archive. These factors make the aleatoric component of the penetration label non-negligible. A model that returns only a conditional mean does not convey whether that mean is reliable for a specific condition, and for risk-screening purposes a calibrated predictive interval that can be interpreted alongside the geometric collision criterion is more informative than an overconfident point estimate [21].

1.5 Research aim

Based on the above problem, this thesis aims to develop a data-driven non-reacting liquid-envelope prediction model for pilot diesel spray. The intention is to predict the evolution of the spray envelope and penetration trend from injection-condition and time information, together with an uncertainty-aware description that is suitable for engineering interpretation.

The thesis does not claim to solve real-engine wall wetting or reacting spray-wall interaction in full. Instead, it positions the resulting model as a front-end surrogate for early-stage impingement-risk screening in marine dual-fuel engine development. In other words, the target contribution is a practical engineering tool that narrows the candidate space and supports later, higher-fidelity validation rather than replacing CFD or engine testing.

1.6 Research methods

To support that aim, this thesis uses high-speed visualization of diesel sprays in a room-temperature pressurized-air chamber. The resulting image sequences are processed to extract penetration and liquid-envelope-related geometric features, forming a dataset for later modeling.

On that basis, the study applies data-driven regression to learn the time-varying liquid-envelope behavior from experimental observations. The model is treated as a surrogate function approximator rather than as a new closed-form spray law. Its outputs are then intended for later geometric risk mapping, that is, for estimating whether a given spray condition is more likely to approach a potential wall-impingement region. Throughout this workflow, the extracted boundary is interpreted as an optical liquid-envelope definition under the chosen diagnostic setup, not as a direct absolute truth of the in-cylinder liquid field [10]. To assess whether the surrogate adds value

beyond available analytical approximations, predicted penetration trends are also placed in the context of classical scaling-law references such as the Hiroyasu–Arai two-stage model [13], allowing the data-driven output to be interpreted against a physically grounded baseline.

1.7 Why this workflow is necessary

The workflow in this thesis is structured deliberately rather than as a convenience implementation detail. A direct end-to-end learning setup from raw spray videos to penetration prediction would be computationally expensive, difficult to interpret, and poorly matched to the current archive scale. It would also make it difficult to audit the intermediate choices that matter for this problem, such as calibration, optical boundary definition, target fitting, and uncertainty-aware regression. This concern is consistent with broader scientific-image-processing practice: mature open-source ecosystems such as Fiji/ImageJ, scikit-image, OpenPIV, and PIVlab emphasize scriptable, inspectable, and reusable processing pipelines rather than one-off manual interpretation [37, 46, 33, 44].

There is also a methodological reason not to learn directly from raw intensity volumes. Mie-scattering images do not provide a direct liquid-volume-fraction truth field; instead, they provide an optical boundary definition whose appearance depends on illumination, scattering, background, and post-processing choices [10, 9]. For that reason, the first requirement is to convert raw videos into more stable plume-resolved observables whose physical interpretation is explicit enough for later modeling.

The finite observation window introduces a further complication. In the present setup, the visible radial extent is limited, so some operating conditions produce right-censored penetration traces whose late-time development is not fully observed. If those incomplete trajectories were used naively as direct labels, the learning target would be biased downward in exactly the region where robust extrapolation matters most. A reduced-order fitting stage is therefore not only a smoothing device but also a censoring-aware target-construction step: it supports robust extrapolation beyond the visible window, improves temporal alignment across conditions, and yields targets that are more suitable for uncertainty-aware regression.

1.8 Main contributions

The main contributions of this thesis are as follows:

1. a CUDA-accelerated workflow for converting terabyte-scale Mie-scattering videos into plume-resolved temporal observables and boundary trajectories;
2. a reduced-order fitting strategy for trajectory alignment, robust extrapolation, and mitigation of right-censoring effects in penetration targets;
3. a multi-stage uncertainty-aware learning framework that combines deterministic MSE training, heteroscedastic NLL training, and teacher-guided refinement on partially observed raw trajectories; and

4. a simplified probabilistic impingement-screening method that decouples penetration prediction from downstream geometric collision evaluation.

1.9 Research scope

The scope of this thesis is limited to non-reacting, non-evaporating pilot diesel spray under controlled pressurized-chamber conditions. The focus is on the macroscopic liquid-envelope geometry and its temporal evolution as observed in the optical setup, rather than on a complete in-cylinder combustion process.

Consequently, combustion chemistry, evaporation, dual-fuel ignition coupling, real piston-bowl and cylinder-wall thermal interaction, and real-engine wall-film truth prediction are outside the scope of this work. The proposed model should therefore be understood as an early-stage screening surrogate, not as a final truth model for engine wall impingement.

Moreover, the MLP-based surrogate developed in this thesis is intended primarily for interpolation within the operating-condition range covered by the available training data. For clearly out-of-distribution extrapolation cases, neither the prediction accuracy nor the uncertainty estimate is claimed at the same level. The model should therefore be interpreted as an engineering screening tool within the supported data domain rather than as a universally reliable truth model for arbitrary unseen conditions.

The restriction to non-reacting, non-evaporating conditions is physically motivated as a conservative screening choice. In the early injection phase that determines how far the liquid envelope can reach, the dominant physics are momentum exchange, aerodynamic drag, and primary and secondary breakup — all of which are present in cold-spray conditions. ECN benchmark studies confirm that non-reacting liquid penetration measured under controlled ambient conditions is reproducible across laboratories and correlates with the same governing parameters — ambient density, injection pressure, and nozzle geometry — that appear in engine-relevant scaling laws [30, 10]. In a cold-spray environment, the liquid length reaches its maximum uninhibited extent because evaporation, which shortens the liquid envelope under hot in-cylinder conditions, is absent. Cold-spray penetration therefore provides a conservative upper bound for impingement screening: a condition whose non-reacting envelope does not approach a wall boundary is less likely to impinge under the corresponding engine condition where evaporative shortening would reduce the liquid reach further. The non-reacting scope supports defensible conservative screening rather than rendering the results irrelevant to engine conditions.

1.10 Thesis structure

The remainder of this thesis is organized as follows. The literature review and technical rationale situate the work in diesel spray imaging, optical diagnostics, and surrogate modeling. The research materials and methods section describes the experimental data, image-processing workflow, feature extraction, trajectory fitting, and model-training procedure. The results section reports the archived processing assets and model

evidence. The conclusion summarizes the current outcome, the main limitations, and the remaining development needs.

2 Literature Review and Technical Rationale

2.1 Why penetration-oriented observables are used in Mie imaging

Mie scattering is useful for visualizing dense liquid sprays because it offers high temporal resolution and comparatively simple experimental implementation [23]. At the same time, the recorded intensity field is not a direct measurement of fuel mass or liquid volume fraction. The scattering response depends on droplet size, refractive index, wavelength, and viewing geometry, while dense regions introduce multiple scattering and shadowing [2, 23]. Because of those effects, direct inversion from pixel intensity to a full physical spray state is ill-posed.

For that reason this thesis adopts the same practical stance used in much of experimental spray analysis: intensity is treated as a carrier of geometric cues, not as a direct thermodynamic field. Operational observables such as penetration, cone angle, projected area, and boundary location are therefore more defensible than claims about exact liquid distribution [23, 22]. In diesel spray research, penetration and cone angle remain standard descriptors precisely because they are interpretable and comparatively robust across controlled experimental campaigns [8, 13, 30].

This emphasis is also consistent with prior optical spray studies. Two-dimensional Mie-scattering work inside optical diesel engines already reported jet-tip penetration and cone-angle information as the practical outputs of image analysis rather than attempting to reconstruct a complete liquid-state field [29]. More recent imaging studies still organize liquid-spray comparison around penetration and spreading-angle measurements, even when the instrumentation and synchronization become more sophisticated [18]. The literature therefore supports the idea that a thesis based on Mie videos should be evaluated primarily by the quality and robustness of its extracted observables.

It is worth noting that the dataset used in this thesis was not chosen by optimizing over available diagnostic modalities but was dictated by the experimental source: the data come from top-view Mie-scattering experiments on a multihole nozzle, even though the same laboratory also has Schlieren, diffuse back-illumination (DBI), and chemiluminescence imaging capability. In practice, each diagnostic has its own physical sensitivity: Schlieren responds to density gradients, DBI measures path-integrated optical thickness, and chemiluminescence indicates the reaction zone. Mie scattering is sensitive to forward and side scattering from liquid droplets, and thus provides high contrast at the liquid-phase boundary, making it well suited for extracting penetration depth and cone angle [23].

Mie imaging also exhibits a characteristic multiple-scattering effect: in the dense liquid-core region near the nozzle, multiple scattering and internal shadowing elevate the local intensity while producing little sharp directional structure, resulting in a

diffuse halo rather than a clean plume boundary [2, 23]. The present approach suppresses this effect through Sobel high-pass filtering: the halo contributes to the slowly varying background intensity field but contributes very little to the local gradient field. Gradient-domain processing therefore acts as a first-order rejection of the multiple-scattering halo without requiring explicit physical modeling of the scattering regime. For the optical path, this thesis adopts a line-of-sight integration simplification: the recorded Mie signal is treated as a path-integrated quantity from a single focal plane. Although the true plume geometry has an umbrella angle and is not strictly planar, this assumption is appropriate for the present goal of macro liquid-envelope characterization rather than three-dimensional liquid-phase reconstruction.

Classical spray-penetration modeling also explains why penetration is such a central descriptor. In the widely used Hiroyasu–Arai picture, the tip penetration is written in stage-wise form with one branch before breakup and another after breakup. In simplified proportional form,

$$S_{\text{tip}}(t) \propto \begin{cases} t, & t < t_b, \\ t^{1/2}, & t \geq t_b, \end{cases}$$

where t_b is the breakup time and the prefactors depend on pressure difference, nozzle diameter, and fluid properties [13]. This type of model is attractive because it gives an interpretable low-order law for the gross penetration history. However, the more recent start-of-injection transient analysis by Zhou et al. shows that the early-time behavior can be more complicated than the classical two-stage law when sac pressurization and needle-opening transients are important. In that study, the SOI stages were found to scale approximately as $t^{3/2}$ initially, then $t^{3/4}$, and only later approach the more familiar $t^{1/2}$ behavior [51]. That observation is directly relevant to the present thesis: pilot diesel injections are short enough that the transient early stage occupies a non-negligible part of the observed signal, so a single rigid empirical law is not always an adequate descriptor of the full trace.

End-of-injection studies extend the same idea further by showing that the relevant time variable itself can change across regimes. In the extended formulation reported by Zhou et al. for tip and tail penetration after EOI, the pre-transition branches are still written against the ASOI clock t , but the decelerating branches include terms in $(t - t_i)^{1/2}$ and $(t - t_i)^{1/4}$, where t_i denotes the injection duration [52]. Their combined expression for the liquid length scale $S_L(t)$ can be written as

$$S_L(t) = \begin{cases} K_v \left(\frac{2\Delta P}{\rho_f} \right)^{1/2} t, & 0 < t \leq t_b, \\ K_p \left(\frac{\Delta P}{\rho_a} \right)^{1/4} d_n^{1/2} t^{1/2}, & t_b < t \leq t_i, \\ K_p \left(\frac{\Delta P}{\rho_a} \right)^{1/4} d_n^{1/2} t^{1/2} - k d_n^{1/2} (t - t_i)^{1/2} (\tan \theta)^{-1/2} \rho_a^{-1/4}, & t_i < t \leq 2t_i, \\ 2^{1/2} K_p \left(\frac{\Delta P}{\rho_a} \right)^{1/4} d_n^{1/2} t_i^{1/4} (t - t_i)^{1/4} - k d_n^{1/2} (t - t_i)^{1/2} (\tan \theta)^{-1/2} \rho_a^{-1/4}, & 2t_i < t. \end{cases}$$

In simplified notation, this means that one part of the penetration law is indexed by elapsed time since the beginning of injection, while another part is indexed by elapsed time since the end of injection or, more precisely, since the decelerating entrainment-wave state starts to matter at the tip. That distinction is useful in the present thesis because it clarifies why the power-law exponent alone is not the full story: the effective time origin also carries physical meaning. It also motivates a softer engineering parameterization, because in short pilot injections the data window may contain both clocks at once and the precise switching instant is not directly observable from a noisy Mie-derived trace.

The breakdown of quasi-steady empirical models is not caused solely by fluid-dynamic staging; it is also compounded by the dynamic behavior of the upstream injector hardware. In a common-rail injection system, the needle-valve opening and closing events are finite-time mechanical transients: before the needle is fully lifted, the effective flow area through the nozzle varies with time, directly affecting the instantaneous injection rate and the near-nozzle flow field [39, 13]. Under high injection pressures, cavitation inside the nozzle can further alter the internal jet structure and affect atomization characteristics in ways that classical penetration correlations do not explicitly capture [2]. Both effects are especially pronounced for short pilot injections, where the needle-opening and closing transients can occupy a substantial fraction of the total injection event, and the quasi-steady assumption holds for only a narrow middle window, if at all. Data-driven models that are not committed to a specific model-form therefore have a practical advantage over rigid power-law correlations when fitting short-injection traces: they do not require the transient regimes to be identified and parameterized in advance, and they can accommodate the combined effect of multiple overlapping physical mechanisms without constructing a piecewise rule for each.

2.2 Supporting theories from computer vision and signal processing

The implemented image-processing chain uses manual calibration, log-domain foreground construction, gradient-based enhancement, polar aggregation, spectral alignment, inverse warping, and binary-shape measurement. Those steps are not arbitrary heuristics; each one is supported by well-established ideas from computer vision and signal processing.

Manual nozzle calibration is justified because the downstream angular analysis assumes a fixed injector center in image coordinates. The calibration task is a noisy circle-fitting problem, for which least-squares circle estimation is a standard tool [36, 43]. Once the center is known, the multihole spray can be described relative to a physically meaningful origin instead of by unstable image coordinates that change with cropping or glare.

The log-ratio preprocessing stage is conceptually related to homomorphic signal processing. When image formation contains multiplicative components, applying a logarithm converts multiplicative variations into additive ones, after which background

subtraction and filtering become more stable [34, 11]. The implementation in this thesis does not claim to be a full homomorphic filter in the classical sense, but it follows the same theoretical motivation: illumination and gain variations should be attenuated before plume geometry is measured.

After foreground construction, the pipeline emphasizes edges and suppresses diffuse low-frequency background structure. Gradient-magnitude operators such as Sobel filters are standard tools for emphasizing boundary transitions, while percentile scaling, automatic thresholding, and morphology are standard ways to stabilize segmentation under non-uniform contrast [11, 41, 50]. This is particularly relevant for Mie videos, where the boundary of interest is often visible first as a change in local contrast rather than as a globally consistent absolute intensity level.

The angular-density stage can be interpreted as a polar-coordinate resampling of the image around the nozzle center. In computer-vision terms, this converts a repeated directional structure in the 2D image into a one-dimensional signal on the angular axis, which is easier to compare across frames and recordings. Once that angular signal is available, using its harmonic structure for alignment is a standard signal-processing idea: periodic patterns are naturally revealed in the Fourier domain, and the phase of the plume-count harmonic provides a compact estimate of rotational offset [6]. Likewise, the subsequent remapping stage follows standard inverse-warping theory, in which output pixels query source coordinates through an inverse transform so that the resampled strip is dense and free of forward-mapping holes [41].

The reduced-order penetration fit used later in the workflow is also grounded in standard estimation theory. When a parametric curve $S(t; \theta)$ is used to summarize noisy observations y_i , the natural formulation is nonlinear least squares, possibly with a robust penalty when a few samples are contaminated by segmentation noise, early-time ambiguity, or truncated trajectories. Huber’s loss is a classical compromise between quadratic least squares and absolute-error fitting: it behaves like squared error near zero residual and like absolute error for large residuals, which makes it less sensitive to outliers while preserving smooth optimization [14, 32]. This directly matches the role of `fit_raw_data.py`, which uses a robust nonlinear fit to convert noisy plume traces into compact parameters for the later learning stages.

These theories support the design choice made in this thesis: before any downstream modeling, the raw video should be transformed into a coordinate system and signal representation in which the plumes are easier to compare, threshold, and measure consistently.

2.3 High-throughput scientific imaging and GPU suitability

The video archive in this thesis is not a set of isolated images but a collection of dense spatiotemporal tensors. Before plume alignment, each recording is naturally represented as a video cube $V \in \mathbb{R}^{F \times H \times W}$. After rotation, cropping, masking, and plume extraction, the same information is reorganized into plume-aligned tensors $I \in \mathbb{R}^{P \times F \times H \times W}$. Most operations applied to those arrays are highly repetitive: background subtraction, local filtering, gradient evaluation, geometric remapping, angular binning, thresholding, binary cleanup, and cumulative-support calculations

are all repeated at many independent pixels, bins, or frames.

This structure is important computationally. Graphics processors are especially well matched to workloads dominated by regular dense-array operations, stencil-like neighborhood computations, and repeated map-reduce style kernels, because the same arithmetic pattern is evaluated over many data elements in parallel [31]. The individual operators used here are standard image-processing primitives [11, 41]; the engineering issue is that, once applied to terabyte-scale 12-bit video archives, their practical cost dominates the full workflow unless they are organized as bulk data-parallel computations.

For that reason, the use of GPU acceleration in this thesis is not treated as a mere implementation preference. It is part of the methodological design. A serial frame-by-frame workflow can make even simple filtering and remapping steps too slow and fragile for iterative model development, whereas a tensor-oriented CUDA pipeline turns the same processing chain into something that can be rerun repeatedly as calibration choices, target definitions, and model stages evolve. In that sense, high-throughput processing is what changes the workflow from practically non-iterable to practically iterable.

2.4 Censoring, truncation, and stable target construction

The finite field of view in optical spray imaging creates a statistical problem as well as a geometric one. When the true late-time penetration extends beyond the visible window, the recorded trajectory is not simply noisy; it is right-censored by the observation limit. In statistics, this is closely related to the broader class of limited or censored dependent-variable problems, where the observed value is only a truncated representation of the latent quantity of interest [45, 3]. Although both references originate in an econometric context, the core statistical structure — a dependent variable that can only be observed below a threshold, with information beyond that threshold suppressed — is formally identical to the situation in optical spray imaging, where the true penetration exceeds the field-of-view boundary and the apparent terminal value is an observation limit rather than a physical one.

Figure 1 shows a representative spray image from the present workflow. Several plumes have already extended beyond the top of the observation window and leave only blurred ring-like edge traces near the upper boundary, whereas other plumes remain fully inside the visible domain. The example makes clear that the apparent terminal penetration in the image can be determined by the camera window rather than by the true spray evolution.

This distinction matters directly for surrogate-model training. If a penetration trace reaches the measurement boundary and the visible endpoint is treated naively as an ordinary regression label, the target is biased downward precisely in the region where long-range penetration should still be increasing. Under that setup, the learner is encouraged to interpret disappearance from the image as physical saturation of the spray. In practice, that kind of label construction can create unstable late-time gradients and artificial flattening when censored and uncensored trajectories are mixed in the same training set.

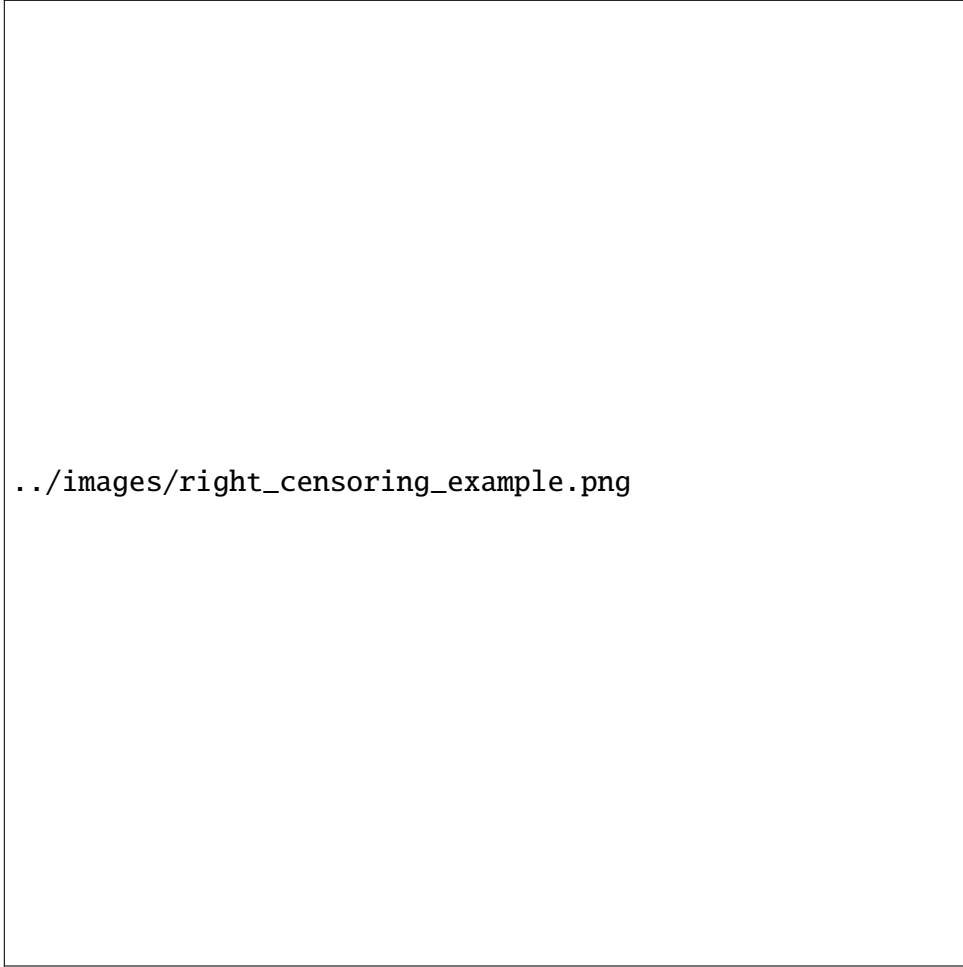


Figure 1: Representative example of right-censoring in the spray images. Some plumes have already extended beyond the upper observation boundary and appear only as blurred ring-like edge traces near the limit, while other plumes are still fully contained within the visible window.

For that reason, the reduced-order fitting stage in this thesis is not introduced only as a smoothing device. It is also a censoring-aware target-construction step. The fitted curve provides a compact and more stable representation of the trajectory, supports conservative extrapolation beyond the visible window, and reduces the sensitivity of downstream learning to the exact point at which the optical trace becomes incomplete. This role is especially relevant for short pilot injections, where start-of-injection and end-of-injection transients already complicate the raw signal shape [51, 52].

2.5 Multihole plume registration and symmetry exploitation

Multihole injectors introduce a problem that is more specific than generic spray segmentation. The image does not contain one isolated jet but several plumes arranged around a common nozzle center, with a known multiplicity and an approximately

repeated angular structure. At the same time, multihole behavior is not identical to an arbitrary repetition of single-hole behavior, because the internal sac flow, hole interactions, and plume-to-plume disturbances can alter the resulting spray morphology [16, 28]. The computational problem is therefore not only to detect spray pixels, but to register repeated plumes consistently across frames and operating conditions.

This is why the thesis moves from Cartesian image coordinates to nozzle-centered polar and plume-aligned coordinates. Once the spray is described on the angular axis, the repeated structure induced by the injector hole pattern becomes much easier to analyze, because plume occupancy can be summarized as a one-dimensional periodic signal rather than as fragmented edge content in the original image plane [41]. The subsequent use of harmonic analysis is then not an arbitrary signal-processing trick, but a symmetry-exploiting registration rule: the known injector multiplicity provides a natural angular frequency whose phase can be used for rotational alignment [6].

2.6 Metric definition and boundary extraction in spray imaging

Fuel-imaging literature also shows that the numerical definition of a spray metric matters. Magnotti and Genzale demonstrated that different liquid-penetration metrics can lead to materially different conclusions when validating diesel spray models, because different observables respond to different aspects of the underlying liquid distribution [26]. This is directly relevant to Mie processing: a reported “penetration” value is not a universal truth, but the output of a measurement rule applied to an optical signal.

The same caution applies to cone angle and boundary extraction. Johnson et al. showed that conventional threshold-based cone-angle estimation from back-scattered diesel images is sensitive to the chosen intensity threshold, and they proposed a profile-based alternative specifically to reduce this subjectivity [17]. That observation supports the present thesis in two ways. First, it justifies treating boundary metrics as operational geometric descriptors instead of as exact liquid-envelope measurements. Second, it motivates the use of a cumulative-distribution penetration front and multiple boundary proxies, because robust metrics should depend on the bulk support of the plume rather than on one fragile leading pixel or one globally fixed threshold.

The result is a workflow that does not attempt to “recover physics” from intensity. Instead, it uses image-processing theory and prior spray-imaging evidence to construct reproducible, physically interpretable geometric observables whose limitations are stated explicitly.

The method-dependence of spray measurements is also systematically documented in the Engine Combustion Network (ECN) benchmark work. By testing nominally identical injectors (Spray A and Spray D) across multiple institutions, ECN found measurable inter-facility variation in liquid-penetration values and explicitly concluded that optically measured liquid boundaries should be understood as quantities defined by a particular measurement protocol, not as absolute ground-truth representations of the liquid phase [10, 9, 27]. The experimental context in this thesis differs from ECN Spray A (high-temperature, high-pressure evaporating conditions) and Spray D (multihole nozzle) — here the data come from a room-temperature pressurized

chamber with a multihole marine nozzle and without fuel evaporation — but the ECN methodological principle applies equally: the numerical values of reported penetration and boundary metrics depend on the post-processing rules, and that dependency must be stated explicitly when results are interpreted.

2.7 Why deterministic, uncertainty-aware, and teacher-guided learning stages are all useful

Data-driven surrogate models have demonstrated clear viability for predicting spray macroscopic characteristics in recent literature. Neural-network approaches have been used to map operating-condition parameters directly to spray tip penetration and cone angle without requiring a full CFD computation as an intermediate step, with several studies reporting prediction accuracy approaching 95% [19, 24]. These results confirm that neural networks are a practically feasible modeling layer for spray macroscopic properties. However, those studies typically assume complete, fully observable penetration trajectories as training labels and do not model inter-condition or inter-cycle uncertainty explicitly. The constraints in the present thesis are more restrictive on all three counts: the short pilot-injection traces are dominated by SOI and EOI transients, the limited field of view produces a large fraction of right-censored trajectories, and the plume-to-plume and shot-to-shot variance is itself engineering-relevant information rather than irreducible noise. The learning framework here therefore extends general data-driven spray surrogates with three design elements that address those specific constraints: censoring-aware target construction, heteroscedastic uncertainty quantification, and a distillation-based refinement stage.

A deterministic multilayer perceptron is a reasonable first surrogate for penetration because it can map a small set of operating-condition features and time to a continuous trajectory [1]. In the archived baseline notebook, the network uses physically scaled inputs and smoothness-related penalties to regularize the trajectory shape.

An uncertainty-aware extension is also motivated, because different operating points and extracted trajectories do not carry the same level of confidence. Heteroscedastic Gaussian negative log likelihood is a standard way to let a network predict both a mean and an input-dependent variance [21]. In regression terms, this corresponds to the conditional model

$$y = \mu(x, t) + \varepsilon(x, t), \quad \varepsilon(x, t) \sim \mathcal{N}(0, \sigma^2(x, t)),$$

which should be read here as a pragmatic statistical assumption rather than as a claim that the spray physics are literally Gaussian at every instant [1, 21]. The present motivation is that the observed penetration spread is the aggregate effect of many unresolved factors: injector-hydraulic transients during pilot injection, sac and needle-opening dynamics near SOI, cycle-to-cycle differences in delivered momentum, synchronization offsets, and residual image-processing noise from the optical measurement chain. Zhou et al. already show that short-injection penetration behavior is strongly transient and regime-dependent near SOI [51]; in the current project, those unresolved perturbations are not modeled mechanistically one by one, so

it is reasonable to treat their combined effect as an input-dependent residual distribution whose variance changes with operating condition and time. In this thesis that idea is treated as a partially completed extension rather than as a finished deployed model, because the archived stage-2 artifacts are incomplete in their current stored form.

The Gaussian output assumption carries an additional computational advantage in the distillation stage. When both teacher and student outputs are represented as univariate Gaussian distributions, the Kullback–Leibler divergence between them has a closed analytical form:

$$D_{\text{KL}}\left(\mathcal{N}(\mu_1, \sigma_1^2) \parallel \mathcal{N}(\mu_2, \sigma_2^2)\right) = \log \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2},$$

which allows the distillation loss to be computed exactly, without Monte Carlo sampling, and therefore without introducing additional gradient variance during training. Regarding the direction of the divergence: forward KL (student approximating teacher) is mean-seeking and distributes mass broadly, whereas reverse KL is mode-seeking and concentrates on the teacher’s main peak with less influence from sparse outlier trajectories. In practice both directions give comparable results in this application, and forward KL was chosen for consistency with the negative log-likelihood training used in Stages 1 and 2 [1, 21]. Knowledge distillation for regression under data scarcity has been shown to reduce student prediction error substantially in regimes where the original training set is incomplete, because the teacher provides stable soft targets in regions where direct labels are absent [20].

A further refinement stage is justified once the supervision becomes uneven over time. Knowledge distillation was introduced as a way to transfer the predictive structure of a teacher network to a student by means of softer targets than hard labels [12]. Closely related teacher-student formulations are also used when supervision is limited or incomplete, because the teacher can provide a stable target signal in regions where direct labels are sparse, noisy, or absent [42]. In the present thesis, that logic is adapted to partially observed penetration trajectories: where uncensored raw observations remain trustworthy, they should directly refine the student; where the late-time raw signal becomes unreliable because of censoring and sample dropout, the teacher continuation is a more conservative guide than treating the disappearing tail as a hard physical label. This is why the third learning stage is methodologically different from simple fine-tuning: it combines direct raw supervision with teacher-guided regularization so that the model can improve early-time fidelity without being pulled downward by incomplete late-time observations.

Taken together, the literature reviewed in this section establishes the following foundations for the present work: Mie scattering theory maps optical intensity to geometric observables in a defensible way; computer vision and signal processing theory supports each step of the image-processing chain; diesel spray physics defines the physical meaning and limits of penetration and cone angle as descriptors; ECN benchmark studies establish the principle that optically defined metrics are method-dependent and must be reported as such; and recent data-driven work confirms that neural-network surrogates can reproduce spray macroscopic behavior at practical

Artifact	Main role	Primary output
GUI.py	Manual nozzle-center calibration	Center, radius, and angular offset for a video set
main.py	Batch preprocessing and feature extraction	Plume-wise penetration traces, binary metrics, and optional boundary CSVs
MLP/fit_raw_data.py	Cleaning, delay alignment, and reduced-order fitting	Cleaned trajectories, flagged trajectories, fitted parameters, and fit plots
MLP/median_penetration_MSE.ipynb	Deterministic baseline training	Stage-1 training logs, scaler state, checkpoint, and loss curves
MLP/median_penetration_NLL.ipynb	Uncertainty-aware extension	Stage-2 training logs, scaler state, and archived uncertainty outputs
MLP/toy_inference_from_run.ipynb	Prototype inference and visualization	Mean and standard-deviation prediction curves plus simple geometry overlays

Table 1: Implemented workflow documented in this thesis.

accuracy. What the existing literature does not address in combination is: (1) a pilot injection regime in which SOI and EOI transients dominate the observable window; (2) training targets that are systematically right-censored by a limited field-of-view chamber; (3) a per-plume extraction pipeline designed for multihole nozzle geometry; and (4) an integrated multi-stage learning framework that couples deterministic baseline, heteroscedastic uncertainty quantification, and distillation-based refinement. The present thesis addresses those four constraints as a unified design, implementing an end-to-end workflow from terabyte-scale Mie video to engineering-ready surrogate output.

3 Research Material and Methods

3.1 Workflow overview

Table 1 summarizes the implemented workflow and the role of each project artifact.

3.2 Calibration in GUI.py

The manual calibration workflow in GUI.py consists of three user-facing stages. First, the operator opens a representative frame in the main annotation interface, adjusts the display settings, and prepares the nozzle-centered view that will be used for the batch anchor. Second, the dedicated calibration dialog is launched, where the user

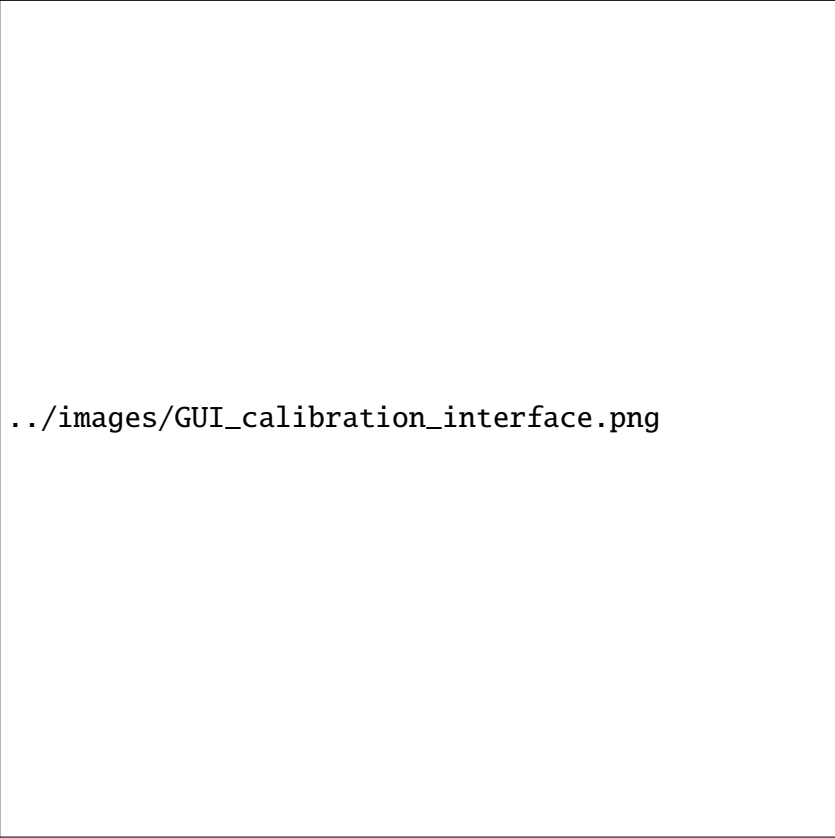


Figure 2: Main annotation interface of GUI .py before calibration. The operator loads a representative frame, adjusts the display, and prepares the manual calibration task.

clicks nozzle-rim points on the selected frame. Third, the fitted center and radius are returned to the main interface, where the recovered geometry can be inspected together with the plume guides and ring overlay before the configuration is saved for later batch processing.

The point-selection dialog enforces at least three clicks and computes the circle through `calc_circle` once the required number of points has been entered. The fitted circle provides the nozzle center, calibration radius, and angular offset for the full recording.

This design is appropriate for the current datasets because the camera and injector geometry stay fixed within a recording. Calibrating once and reusing the same center for all frames avoids frame-to-frame jitter in the downstream plume-angle analysis.



`../images/GUI_calibration_interface.png`

Figure 3: Calibration point-selection dialog in `GUI.py`. The user marks nozzle-rim points that are later used for circle fitting and angular-offset estimation.

3.3 Batch video processing in `main.py`

The file processor in `main.py` is the operational center of the image-analysis workflow. For each `.cine` recording, `_process_cine_file` loads the video to the active array backend, reuses an annular ring mask for the whole folder, calls `mie_multihole_preprocessing`, calls `mie_multihole_postprocessing`, extracts plume-wise penetration curves, computes boundary-based geometric metrics, and finally exports synchronized CSV and metadata files. The important engineering point is that the numerically heavy part of this chain stays backend-local: when CuPy is available, convolution, angular binning, remapping, thresholding, connected-component cleanup, and morphology all remain on the CUDA path until the final tabular export.

This implementation choice was motivated by the practical limits of the earlier in-house route, which relied on frame-by-frame MATLAB processing without effective parallel acceleration. The criticism is not directed at MATLAB as a language: PIVlab, for example, is a mature MATLAB-based open-source PIV toolbox [44]. Rather, the limitation was the local batch architecture. In that form, processing about 50 GB of video could take roughly a day and still remain prone to crashes. Moving the heavy image operations into a bulk CUDA-oriented pipeline is therefore treated here as part



Figure 4: Calibration result shown in the main interface after circle fitting. The recovered nozzle center, guide directions, and radial bounds provide the geometric reference used by the downstream batch pipeline.

of the method rather than as a cosmetic software optimization.

3.3.1 Calibration hand-off and batch anchoring

The batch stage starts from the calibration produced in Section 3.2. In `calc_circle`, the clicked nozzle-rim points (x_i, y_i) are fitted with the algebraic circle model

$$x_i^2 + y_i^2 + Dx_i + Ey_i + F = 0,$$

solved in least-squares form by `np.linalg.lstsq`. The recovered parameters are

$$c_x = -\frac{D}{2}, \quad c_y = -\frac{E}{2}, \quad r = \sqrt{c_x^2 + c_y^2 - F},$$

and the same routine also estimates an average angular offset from the clicked rim points relative to an ideal equally spaced set. `GUI.py` writes these values to `config.json`, after which `main.py` can process a full folder without manual re-centering. This one-time calibration is deliberately user-facing rather than fully automatic: the operator can zoom, choose a representative frame, and anchor the full batch with a physically meaningful nozzle center [36, 43].

3.3.2 Bulk and robust preprocessing

The first algorithmic contribution inside the batch pipeline is a foreground construction stage designed for noisy Mie videos rather than for idealized binary images. In `refined_log_subtraction`, the raw frame $V_t(x, y)$ is mapped to

$$L_t(x, y) = \log(V_t(x, y) + \varepsilon), \quad L_{\text{bkg}}(x, y) = \text{median}_{t < t_{\text{SOI}}} L_t(x, y),$$

followed by the log-ratio reconstruction

$$R_t(x, y) = \exp(L_t(x, y) - L_{\text{bkg}}(x, y)) - 1 = \frac{V_t(x, y) - B(x, y)}{B(x, y)}.$$

The implementation then multiplies R_t by an intensity gate $M_t(x, y) = \mathbf{1}[V_t(x, y) > \alpha B(x, y)]$, subtracts a pre-injection baseline, applies a soft threshold

$$F_t(x, y) = \max(M_t(x, y)R_t(x, y) - R_0(x, y) - \tau, 0),$$

and rescales the result by robust quantiles before any geometric processing. This sequence is important for Mie videos because the raw intensity is not a direct measure of liquid volume fraction, and because the practical objective is stable geometry extraction under non-uniform illumination, glare, and background texture rather than exact optical inversion [2, 23, 11].

The need for this robust preprocessing can already be seen at the visualization level. Figure 5 compares the same spray frame under two display gains while the remaining GUI parameters are held fixed. At the lower gain, the plume cores remain comparatively compact. At the higher gain, a bright halo spreads around the spray core and contaminates a much larger region of the image. For the purposes of this thesis, that behavior is treated as practical evidence that Mie-based intensity is strongly affected by multiple scattering, glare, and display-sensitive background contamination. As a result, a naive subtraction from a single grayscale background or a direct threshold on intensity alone would be too sensitive to gain-dependent spreading. This is precisely why the implemented pipeline uses a robust background estimate from pre-injection statistics and then shifts the representation toward edge-dominant Sobel features, whose geometric response is more stable than raw brightness magnitude alone.

3.3.3 Ultra-efficient Sobel high-pass on the CUDA backend

The second contribution is the explicit edge-enhancing high-pass stage applied directly to the full video cube. In `arr_3d_sobel_magnitude_cupy`, Sobel kernels K_x and



Figure 5: Same spray frame shown in `GUI.py` at gain values 1.0 and 7.0, with the remaining display parameters unchanged. The higher-gain view makes the bright halo around the spray much more apparent, illustrating how multiple scattering, glare, and diffuse background contamination can distort raw intensity-based interpretation.

K_y are generated once and applied with `convolution_2D_cupy`:

$$S_t(x, y) = \sqrt{(K_x * F_t)^2 + (K_y * F_t)^2}.$$

A subtler efficiency comes from the rank-1 structure of the Sobel kernel itself. In `build_2d_kernel`, each directional kernel is assembled as an explicit outer product,

$$K_x = \mathbf{g} \otimes \mathbf{d}_g, \quad K_y = \mathbf{d}_g \otimes \mathbf{g},$$

where $\mathbf{g}$ is a discrete Gaussian envelope and $\mathbf{d}_g$ is its derivative. An outer-product matrix has algebraic rank exactly one. The function `_separable_factors_cupy` tests for this automatically: it computes a truncated SVD of the kernel matrix, $K \approx \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top$, and if the second singular value is negligible the 2-D convolution is replaced by two sequential 1-D passes along the spatial axes. For a $k \times k$ kernel this reduces the per-pixel multiply-add cost from $O(k^2)$ to $O(2k)$. Because the Sobel kernel is rank-1 by construction, the separable path is taken unconditionally at runtime, and both K_x and K_y are applied as a pair of 1-D convolutions across the full video cube.

The output is then robust-scaled again and masked by the ring geometry. In implementation terms this matters because it converts a low-contrast grayscale plume into an edge-dominant representation while keeping the whole operation on the CuPy backend. The contribution is not the Sobel operator itself, which is classical, but its integration into a bulk video workflow that avoids repeated host-device copies and prepares every plume for the same downstream geometric treatment [11, 41].



../images/fig_mie_preproc_background.png

Figure 6: Intermediate preprocessing visualization copied from the repository workflow. From left to right: raw frame, estimated background from pre-injection statistics, and log-subtracted foreground.

3.3.4 Angular density as polar resampling

After preprocessing, the pipeline does not search for plume directions by contour tracing in Cartesian coordinates. Instead, `angle_signal_density_auto` performs a polar re-parameterization around the calibrated nozzle center. For every pixel,

$$\theta(x, y) = \text{atan2}(y - c_y, x - c_x) \bmod 360^\circ,$$

and the per-frame angular signal for bin k is accumulated as

$$A_t(k) = \sum_{(x,y) \in \mathcal{B}_k} I_t(x, y), \quad D_t(k) = \frac{A_t(k)}{\#\mathcal{B}_k},$$

where $\mathcal{B}_k$ is the set of pixels whose angle falls into bin k . In the CuPy implementation, the binning and weighted sums are performed directly on GPU arrays by `cp.digitize`

../images/fig_mie_preproc_sobel.png

Figure 7: Repository-generated preprocessing output showing the foreground image, Sobel magnitude, and the final robust-scaled high-pass representation used by the multihole segmentation stage.

and `cp.bincount`. Conceptually, this is a radial collapse of the 2D image into a 1D angular occupancy signal, which is a much more stable object for multihole alignment than raw edge fragments in the original frame [41].

3.3.5 FFT-based self-adaptive rotation alignment

The plume guide angles are then estimated from the periodicity of the angular signal rather than from a fixed offset. Let

$$\bar{A}(k) = \sum_t A_t(k)$$

be the time-summed angular profile, and let P be the known number of plumes. The function `estimate_offset_from_fft` computes the real FFT of $\bar{A}$, reads the phase

of the plume harmonic P , and converts it to an angular offset

$$\hat{\phi}_0 = -\frac{\arg(\mathcal{F}\{\bar{A}\}[P])}{P} \frac{180^\circ}{\pi}.$$

This is what makes the alignment self-adaptive: the code uses the observed harmonic phase of the current dataset instead of assuming a manually typed reference angle for every recording. `main.py` then constructs plume directions

$$\phi_p = \frac{360^\circ p}{P} - \hat{\phi}_0, \quad p = 0, \dots, P-1,$$

and combines them with a triangle-threshold occupancy mask and short-gap filling to reject empty angular sectors. The FFT itself is standard; the contribution here is using the known injector multiplicity to turn the angular-density signal into an automatic plume registration rule [6, 41].

The same occupied-angle mask also suggests a simple cone-angle proxy. Treat the bin-wise mask as periodic on $[0, 360^\circ)$, merge any `True` runs that cross the $0^\circ/360^\circ$ boundary, and convert each contiguous run length from bins to degrees. The resulting angular widths quantify the support occupied by each plume over the full recording. In the notebook figure workflow we report both the total occupied angle and the mean run width; the latter is used as a coarse cone-angle proxy because it summarizes the average angular spread of a plume without requiring a separate edge fit in every frame.

3.3.6 Inverse affine mapping and plume-strip extraction

Once the target angles are known, each plume is remapped into a common nozzle-left strip coordinate system. The forward affine transform built in `build_rotation_affine` is

$$\begin{bmatrix} u \\ v \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \end{bmatrix} + \mathbf{b}, \quad \mathbf{A} = \begin{bmatrix} \cos \phi_p & \sin \phi_p \\ -\sin \phi_p & \cos \phi_p \end{bmatrix}, \quad \mathbf{b} = \mathbf{t} - \mathbf{A}\mathbf{c},$$

where $\mathbf{c} = (c_x, c_y)^\top$ is the nozzle center and $\mathbf{t}$ places the nozzle at the left edge and mid-height of the output strip. The actual remap uses the inverse relation

$$\begin{bmatrix} x_{\text{src}} \\ y_{\text{src}} \end{bmatrix} = \mathbf{A}^{-1} \left(\begin{bmatrix} u \\ v \end{bmatrix} - \mathbf{b} \right),$$

implemented in `build_affine_inverse_maps_cupy`. For each plume, the map arrays `mapx` and `mapy` are precomputed once, and `remap_video_cupy` samples the full video with nearest, bilinear, bicubic, or Lanczos kernels. Importantly, these maps are not spray-intensity images: the pixel value in `mapx` encodes the source-image x coordinate sampled for that output pixel, and the pixel value in `mapy` encodes the corresponding source-image y coordinate. The multihole pipeline uses a fixed strip size `OUT_SHAPE = (outer_radius/2, outer_radius)` and applies a triangular plume mask after remapping. This inverse-map formulation is computationally efficient and numerically clean, because every downstream plume metric is measured in the same local coordinate system [41].



Figure 8: Processed angular-density outputs from the repository workflow: per-frame angular profiles, time-summed profile with plume guide angles, FFT harmonic content, and the occupied-angle mask used to stabilize multihole alignment.

3.3.7 Robust CDF penetration from aligned plume strips

The penetration estimator is deliberately based on cumulative intensity support instead of on the single brightest pixel or the outermost isolated blob. For each plume strip $I_p(t, y, x)$, `penetration_cdf_all_plumes` first collapses the cross-stream direction,

$$H_p(t, x) = \sum_y I_p(t, y, x),$$

subtracts a left-edge baseline from each frame, clips negative values, and generates a support mask by automatic thresholding, largest-component cleanup, and binary opening. Within that support, `penetration_cdf_front` forms the axial cumulative

../images/fig_angular_mask_comparison.png

Figure 9: Two coordinate-system views of the same spray-occupied angular support accumulated over the full video. Left: the legacy Cartesian mask obtained by projecting the occupied angular bins back into image space around the nozzle center, together with the total occupied angle and the mean run-width cone-angle proxy. Right: the polar occupied-angle mask used directly in the current workflow, annotated with the angular width of each contiguous occupied segment.

distribution

$$C_p(t, x) = \frac{\sum_{\xi \leq x} \tilde{H}_p(t, \xi)}{\sum_{\xi} \tilde{H}_p(t, \xi) + \varepsilon},$$

and defines the penetration front as the first location satisfying

$$\hat{x}_p(t) = \min\{x : C_p(t, x) \geq q\},$$

with q close to one. The code then enforces monotonicity by `np.maximum.accumulate`, subtracts the inner radius to express distance from the nozzle annulus, and in `main.py` discards traces that show non-physical backward jumps or only very short valid runs.



Figure 10: Repository-generated visualization of the inverse affine remapping stage. Top-left and top-right: `mapx` and `mapy`, whose pixel intensities encode the source-image x and y coordinates sampled for each output-strip pixel rather than spray brightness. Bottom-left: one rotated plume strip with the orifice-local origin marker at $(H/2, \text{inner_radius})$, axial penetration overlays plotted at $\text{inner_radius} + \text{metric}$, and the polar binary-penetration metric drawn as a radius- r arc centered at the orifice. Bottom-right: the triangular plume mask used after remapping.

This CDF front is one of the main practical contributions of the workflow: it is much less sensitive to broken leading-edge pixels than a naive max-index definition, while still producing a penetration quantity aligned with standard diesel-spray usage [8, 13, 30].

3.3.8 Boundary metrics and geometric proxies

Binarized plume strips are also passed to `extract_single_plume_features`, which computes several secondary observables. The projected area is

$$A_p(t) = \sum_{x,y} B_p(t, y, x).$$

Axial binarized penetration is obtained from the rightmost column whose occupancy exceeds a pixel threshold, while polar penetration is defined from the upper and lower boundary point clouds as

$$R_p(t) = \max \left(\max_i \sqrt{x_{u,i}^2 + y_{u,i}^2}, \max_j \sqrt{x_{l,j}^2 + y_{l,j}^2} \right).$$

The code also stores a volume-like geometric proxy,

$$V_p(t) = x_{\text{scale}} \frac{\pi}{4} \sum_x (w_u(t, x) + w_l(t, x))^2,$$

where w_u and w_l are the upper and lower half-widths and $x_{\text{scale}} = 1/\cos((180^\circ - \theta_{\text{umb}})/2)$ compensates for umbrella-angle tilt when needed. Cone-angle proxies are computed in two ways: by averaging local boundary angles

$$\alpha_{\text{avg}}(t) = \overline{\arctan(y_u/x_u)} - \overline{\arctan(y_l/x_l)},$$

and by fitting fixed-intercept boundary lines and taking

$$\alpha_{\text{lg}}(t) = \arctan(m_u) - \arctan(m_l).$$

Near-nozzle opening and closing times are finally inferred from a small detection window close to the injector. These metrics matter because they turn each aligned strip into a compact geometric state vector; however, the downstream modeling in this thesis remains focused on penetration, which is the best-supported target in the archived notebooks [23, 22].

3.4 Trajectory cleaning and reduced-order fitting in `MLP/fit_raw_data.py`

The learned models do not train directly on the raw exported traces. Instead, `fit_raw_data.py` first cleans and time-aligns them. For each plume, the script finds the earliest valid penetration segment, takes the first few positive samples after onset (currently two, controlled by `NUM_POINTS_SOI_LINEAR_REGRESSION`), fits a straight line to those early nonzero points, and uses the x -axis intercept of that line as a subframe opening-delay estimate whenever the fitted slope is positive. If an area-based onset marker is available, that frame anchors the delay scan; otherwise the code falls back to the first positive penetration frame. The resulting raw delay is then clipped to

../images/fig_mie_metrics_summary.png

Figure 11: Examples of quantitative outputs produced after plume alignment and binarization: penetration descriptors, plume size, cone-angle proxies, and a representative maximum projection of the binarized plume strip.

a small median-centered window across plumes so that one anomalous plume does not over-shift the whole group.

That delay estimate is used in two coupled ways. The penetration array itself is shifted left by the nearest integer number of frames, while the aligned time vector retains the fractional residual through the exact offset subtraction. In this way the implementation does not merely discard leading zeros; it performs a plume-onset registration that preserves subframe timing information before the later reduced-order fit. After alignment, the script removes sudden unrealistic jumps, converts pixels to corrected physical length, and flags low-quality fits through a set of rule-based masks.

Each accepted trajectory is then fitted with a smooth two-regime model,

$$S(t) = (1 - w(t)) k_{\text{sqrt}} \sqrt{t} + w(t) k_{\text{quarter}} t^{1/4}, \quad w(t) = \sigma\left(\frac{t - t_0}{s}\right),$$

where k_{sqrt} , k_{quarter} , t_0 , and s are optimized in log space. Concretely, the fitting problem is posed as a robust nonlinear least-squares estimation,

$$\min_{\theta} \sum_i \rho_{\delta}(S(t_i; \theta) - y_i), \quad \theta = [\log k_{\text{sqrt}}, \log k_{\text{quarter}}, \log t_0, \log s],$$

with the Huber penalty

$$\rho_{\delta}(r) = \begin{cases} \frac{1}{2}r^2, & |r| \leq \delta, \\ \delta \left(|r| - \frac{1}{2}\delta \right), & |r| > \delta. \end{cases}$$

In the implementation, this objective is solved by `scipy.optimize.least_squares(..., method="trf", loss="huber")` with $\delta = 1$, while the log-parameterization keeps the fitted gains and transition scale positive [14, 32]. This fit is not introduced as a baseless convenience curve. It is a reduced-order, data-driven adaptation of the regime-change viewpoint used in diesel spray-penetration literature: Hiroyasu and Arai describe spray development in terms of changing penetration behavior across stages [13], and later work on start-of-injection transients shows that short injections are difficult to represent well with one rigid empirical expression over the entire trace [51]. That consideration is especially relevant here because the present experiments use pilot diesel injection, for which the near-SOI portion is short, highly transient, and relatively noisy.

The model should therefore be read as a temporal parameterization rather than as an explicit physics law. Its purpose is twofold: to provide believable extrapolation farther from SOI, where the downstream surrogate needs stable targets, and to provide acceptable interpolation near SOI, where the relative error is larger and the apparent early-time physics are harder to constrain from the image trace alone. The sigmoid gate is important for that role. Instead of imposing a hard breakpoint between two empirical branches, it creates a differentiable transition, which keeps the fitted curve smooth and makes it more compatible with the later neural-network stage. In other words, the early transient physics are not forced into a rigid closed-form law here; they are only approximated well enough that the subsequent network can learn the remaining condition-dependent behavior from the full operating-point inputs.

This interpretation becomes clearer when the fit is compared with the extended EOI expressions in the spray literature. The Zhou et al. post-EOI model distinguishes the global injection clock t from the delayed clock $(t - t_i)$, because the late penetration branch is activated only after the end-of-injection dynamics have propagated downstream [52]. The present pipeline handles that delay in two layers. First, the preprocessing stage estimates a plume-wise onset delay from the early linear extrapolation described above and shifts the observed series toward a common $t \approx 0$ reference. Second, the reduced-order fit itself still does not contain an explicit delayed clock inside the parametric branch. Instead, it uses

$$w(t) = \sigma\left(\frac{t - t_0}{s}\right)$$

to suppress the quarter-root branch before a learned transition time t_0 , and then allows the $t^{1/4}$ term to dominate only after that transition. This is not algebraically identical to

replacing $(t - t_i)^{1/4}$ with $t^{1/4}$. Rather, it is an engineering reparameterization: on a finite observation window, a delayed fourth-root branch can often be represented adequately by a gated unshifted fourth-root branch with adjusted amplitude and transition location, especially when the early part of that branch is strongly attenuated by $1 - w(t)$ or $w(t)$ near zero. In that sense, the delayed physical clock is absorbed into the learned switching parameters (t_0, s) and into the branch amplitudes rather than being imposed as a fixed known constant.

I therefore consider the engineering approach reasonable for the stated objective of this thesis, with one clear boundary. It is reasonable because the fit is used as a smooth compressor of noisy trajectories, not as a final mechanistic spray law; it keeps the curve differentiable for later learning; and it leaves the operating-condition dependence to the neural network instead of forcing every physical effect into the closed-form fit. The boundary is that this justification should be presented as a reduced-order surrogate motivated by empirical penetration theory, not as proof that the shifted-time physics have disappeared. A stricter physics-oriented study would compare the present fit directly against a branch written explicitly in $(t - t_i)$ and report the ablation.

The archived local analysis supports this interpretation. The exploratory notebooks in `MLP/interpretable_test` explicitly describe the model as a Hiroyasu–Arai-inspired two-branch blend, while `fitted_stats_analysis.ipynb` and the archived clean-fit CSV tables show that k_{sqrt} behaves more like an optional shape descriptor than a uniformly stable physical coefficient. Across the current clean-fit archive, the square-root branch amplitude at the fitted transition time is smaller than half of the quarter-root branch for the large majority of accepted rows, so many successful fits effectively degenerate toward a quarter-root-dominated curve. This is acceptable for the intended use: the physical drivers are later given directly to the neural network, while the fit itself remains self-adaptive and physics-light. In the archived production outputs under `MLP/synthetic_data`, 105 594 of 123 902 candidate rows are marked as successful by the raw nonlinear fit, corresponding to a practical success rate of about 85% before the stricter post-fit masking stage. The practical purpose of this fit is therefore not to claim a universal penetration law, but to compress noisy plume trajectories into smooth, differentiable, and trainable targets while still preserving the dataset’s heteroscedastic variation for the later learning stages.

The stricter post-fit masking stage is also an important part of the engineering contribution. In `apply_filter_masking`, the script first forms a far-time penetration proxy by evaluating the fitted curve at $t_{\text{far}} = 5$ ms. The row is retained only if that far-time value remains inside a broad physically plausible window and if the fit parameters pass basic sanity checks on t_0 , s , point count, and fit cost. The code then computes robust grouped outlier scores using the median absolute deviation,

$$z_{\text{robust}}(x) = 0.6745 \frac{x - \text{median}(x)}{\text{MAD}(x) + \epsilon}, \quad \text{MAD}(x) = \text{median}(|x - \text{median}(x)|),$$

applied to t_0 , $\log(1 + \text{RMSE})$, and $\log(1 + \text{cost_per_point})$ within each file group. Rows with $|z_{\text{robust}}| > 3$ on any of those diagnostics are marked as outliers and excluded from the clean archive. This matters for the later neural-network stages because it removes evidently unstable fits before they can distort the training distribution. The

result is a deliberately conservative curation step: the network does not see every fitted row, only those trajectories that remain numerically plausible both near the fitted transition and at a fixed far time.

3.5 Sparse-feature instability in the direct-feature baseline

Before the v2 architecture was finalised, an earlier nine-feature MLP was trained with direct pressure and diameter inputs. A systematic diagnostic documented in `MLP/v1_direct_feature_training/slides_sparse_feature_instability` revealed a structural pathology. Nozzle diameter has only six levels in the dataset and is not uniformly available across pressure combinations: all six diameter levels appear only on the family ($P_{\text{inj}} = 2000$ bar, $P_{\text{cb}} = 4$, $P_{\text{ch}} \in \{5, 10, 15\}$ bar). Injection pressure has four levels. Because a smooth MLP treats these as continuous inputs, it can produce derivative structure along the diameter and pressure axes that the data simply do not constrain. Dense MLP sweeps over diameter showed five to six gradient sign changes with worst-secant-deviation ratios below one and curvature norms of order 10^7 ; injection pressure showed unsupported extrapolation on thinly supported slices. This pathology is a structural consequence of the sparse-feature geometry and persists across time. Adding regularisation would suppress instability at the cost of useful capacity everywhere; the underlying problem is that a smooth MLP is being asked to learn a physically meaningful continuous law from an effectively discrete sample grid. The v2 architecture therefore addresses this at the source.

3.6 Engineered-feature design in MLP/v2_engineered_feature

The v2 surrogate replaces the four weakly supported pressure and diameter inputs with a single hard scaling group derived from the Hiroyasu–Arai post-breakup branch [13]. From the post-breakup form

$$S(t) \propto K_p \left(\frac{\Delta P}{\rho_a} \right)^{1/4} d_n^{1/2} t^{1/2},$$

all dependence on injection pressure, chamber pressure, chamber density, and nozzle diameter collapses to one per-condition length scale

$$A = \left(\frac{\Delta P}{\rho_a} \right)^{1/4} d_n^{1/2},$$

computed in `engineered_feature_common.build_canonical_feature_table` from the test-matrix JSON. For Nozzle1–Nozzle8 the chamber density is taken directly from `chamber_density_kg_per_m3`; for DS300, where only `chamber_pressure_bar` is recorded, ρ_a is reconstructed from the linear law $\rho_a = 5 \text{ kg/m}^3 \times P_{\text{ch}}/4.42 \text{ bar}$ calibrated against the Nozzle1–8 entries.

The four sparse columns are removed entirely, leaving two feature-vector variants. The five-feature `a_only` variant

$$[t_{\text{norm}}, \theta_{\text{tilt},z}, n_{\text{plumes},z}, t_{i,z}, P_{\text{cb},z}]$$

treats A as a target divisor only. The six-feature `a_plus_log_a` variant additionally contains $\log A_z$ as a one-dimensional residual axis. The training target is reparameterised to the non-dimensional penetration $\hat{S}(t) = S(t)/A$; the network predicts $\hat{\mu}$ and $\log \hat{\sigma}^2$, and physical predictions are recovered at inference as $\mu_S = A \hat{\mu}$ and $\sigma_S = A \hat{\sigma}$.

Before any optimisation, `run_pretrain_collapse_check` tests the scaling assumption on the actual dataset: it divides each trajectory by its own A , reports the per-time-slice variance reduction, and computes a linear-regression R^2 of $\hat{S}(t)$ on the kept features. In the archived runs the check passed with a median post-0.5 ms variance reduction of 59.5% (collapse ratio 0.405) and a sparse-feature R^2 dropping from 0.34 on S to 0.14 on $\hat{S}$, confirming that A accounts for the dominant diameter and pressure variation and that the network is learning only the residual structure on the dense, well-supported axes. If the check fails, training aborts by default, making the physical prior falsifiable rather than asserted.

The driver scripts `train_stage1_mse.py` and `train_stage2_nll.py` also introduce group-aware data splits: `assign_splits_by_group` assigns whole `sample_group_id` groups to train, validation, and test, preventing leakage between curves that share an operating point. Stage 2 warm-starts strictly from a Stage-1 engineered run directory and reuses its scaler state, guaranteeing consistent variant choice and A -scaling across both stages.

3.7 Stage-1 MSE baseline on the non-dimensional target

Stage 1 trains a deterministic mean model on a representative-trajectory subset. For each file-level operating-condition group, the fitted curve is evaluated at $t_{\text{far}} = 5$ ms, the group-median penetration is computed, and the nearest row is kept as the stage-1 representative. In the archived run `stage1_engineered_mse_a_only_20260410_020317` this produced 526 representative trajectories, split by group into training, validation, and test sets.

The loss applied to the non-dimensional target $\hat{S}$ is

$$\begin{aligned} \mathcal{L}_{\text{stage1}} = & \text{MSE}(\hat{\mu}, \hat{S}) + \lambda_{\text{var}} \mathbb{E} \left[\left(\log \hat{\sigma}^2 - c_0 \right)^2 \right] \\ & + \lambda_1 \mathbb{E} \left[\text{ReLU} \left(-\frac{\partial \mu_S}{\partial t} \right)^2 \right] + \lambda_2 \mathbb{E} \left[g(t) \text{ReLU} \left(\frac{\partial^2 \mu_S}{\partial t^2} \right)^2 \right], \end{aligned}$$

where $c_0 = -2$ is the log-variance prior centre, $g(t) = \sigma((t - t_c)/\tau_c)$ gates the concavity penalty to activate after the transient region, and the derivative penalties are evaluated on the physical reconstruction $\mu_S = A \hat{\mu}$ so that the monotonicity and concavity constraints apply at the observable scale. The network is a three-hidden-layer MLP with dimensions [512, 512, 128], GELU activations, dropout 0.3,

and AdamW optimisation with gradient clipping and early stopping on validation loss. The run achieved a best validation MSE (on the scaled target) of 18.94 at epoch 115 and a test physical MAE of 8.1 mm. The run directory contains `best_model_stage1.pt`, `scaler_state.json`, epoch and iteration loss logs, and a `pretrain_collapse_summary.png`.

3.8 Stage-2 NLL heteroscedastic extension

Stage 2 warm-starts from the Stage-1 checkpoint and expands the training set to all filtered trajectories, replacing the MSE data-fit term with a heteroscedastic Gaussian NLL on the scaled target:

$$\mathcal{L}_{\text{stage2}} = \underbrace{\mathbb{E} \left[\frac{1}{2} \left(\log \hat{\sigma}^2 + \frac{(\hat{S} - \hat{\mu})^2}{\hat{\sigma}^2} \right) \right]}_{\text{heteroscedastic NLL on } \hat{S}} + \lambda_1 \mathbb{E} \left[\text{ReLU} \left(-\frac{\partial \mu_S}{\partial t} \right)^2 \right] + \lambda_2 \mathbb{E} \left[g(t) \text{ReLU} \left(\frac{\partial^2 \mu_S}{\partial t^2} \right)^2 \right], \quad \hat{\sigma}^2 = \exp(\log)$$

The derivative penalties remain on the physical-space reconstruction $\mu_S = A \hat{\mu}$. The row table expands to 14,594 filtered trajectories, retaining all rows that pass the fit-level masks and a far-time z-score filter on group-wise spread. The split is assigned at the `sample_group_id` level rather than at the individual plume-trajectory level, so trajectories from the same physical case do not leak across train, validation, and test partitions. With 512 time samples per trajectory over 0 ms to 5 ms, the archived Stage-2 table contains 7,472,128 pointwise samples: 10,333 trajectories from 370 groups for training, 2,092 trajectories from 78 groups for validation, and 2,169 trajectories from 78 groups for testing.

The retained trajectories span the dominant operating range in the archive: control back-pressure from 1 to 4 bar, injection pressure from 1,400 to 2,200 bar, injection duration from 380 to 1,200 μs , and fitted far-time penetration from 58.1 mm to 202.6 mm. The Stage-2 model therefore does not only learn a single median curve; it sees the remaining within-condition plume and repeat variation after outlier removal. The heteroscedastic output is used to separate two effects that would be conflated by a pure MSE objective: the conditional mean trajectory and the conditional spread around that mean.

Training reuses the Stage-1 feature mapping and scaler state. The input vector remains `[time_norm_0_5ms,  $\theta_z$ ,  $N_{\text{plume},z}$ ,  $\Delta t_{\text{inj},z}$ ,  $p_{\text{back},z}$ ]`, the architecture remains the three-hidden-layer MLP with dimensions `[512, 512, 128]`, and the network is warm-started from `best_model_stage1.pt`. The Stage-2 run uses AdamW with learning rate 8×10^{-4} , weight decay 10^{-4} , batch size 96, gradient clipping at norm 1.0, log-variance bounds `[-10, 6]`, and early stopping with patience 35. Thus the only intentional change in the data-fit term is the replacement of deterministic MSE by the Gaussian NLL and its learned variance output.

Table 2 shows that the best validation NLL is reached at epoch 66, while early stopping terminates the run at epoch 101 after no better validation loss is found. The held-out test MAE of 9.18 mm is close to the validation MAE, and the mean predicted standard deviation of 11.44 mm remains larger than the point-error scale. This is

Record from epoch_loss.csv	Epoch	Pointwise samples	Loss / scaled NLL	Physical MAE	Mean predicted std.
Train at best validation epoch	66	5,290,496	2.156 / 2.148	9.83 mm	12.27 mm
Validation, best checkpoint	66	1,071,104	2.038 / 2.038	8.81 mm	11.71 mm
Validation, stopping epoch	101	1,071,104	2.053 / 2.053	8.97 mm	12.11 mm
Test, restored best checkpoint	101	1,110,528	2.083 / 2.083	9.18 mm	11.44 mm

Table 2: Epoch-level Stage-2 loss summary from MLP/runs_mlp/stage2_engineered_nll_a_only_20260410_020607/epoch_loss.csv. The test row is written after early stopping, but the model weights are restored to the best validation checkpoint before test evaluation.

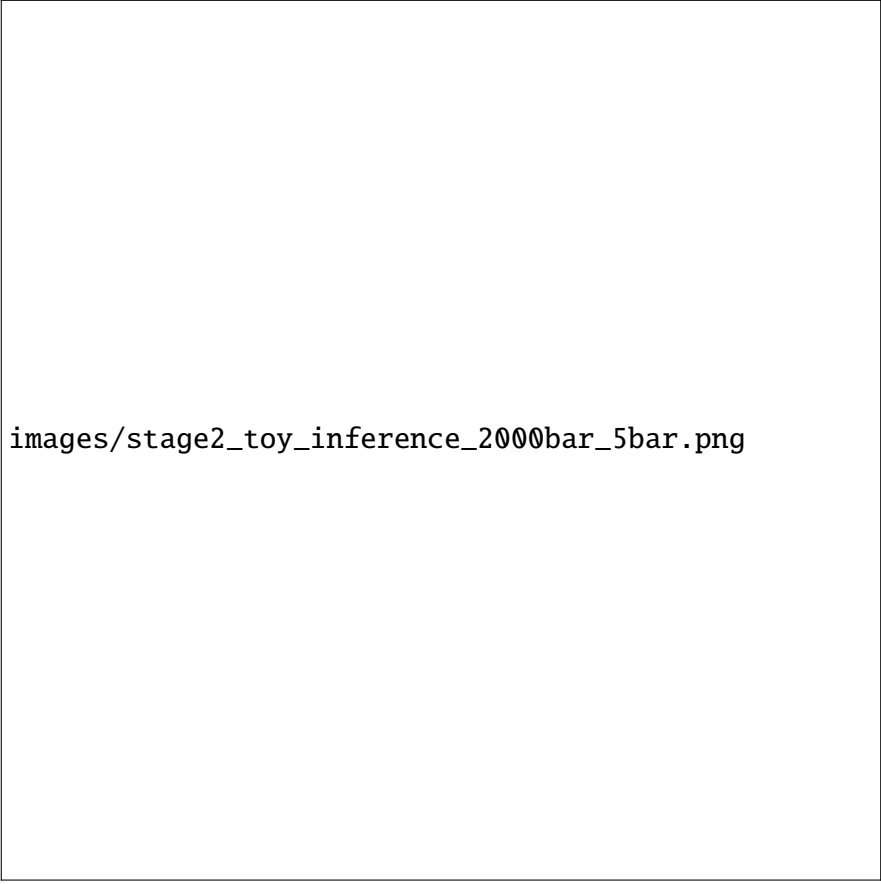
the desired behaviour for this stage: the model is not forced to explain all residual plume-to-plume variation through the mean function, but instead assigns a nonzero conditional uncertainty that can be used by the later impingement and distillation stages. The run directory contains `best_model_stage2.pt`, `scaler_state.json`, `row_table.csv`, `epoch_loss.csv`, `iteration_loss.csv`, `test_summary.csv`, and `stage2_loss_curves.png`; `best_model_stage2.pt` serves as the teacher checkpoint for the subsequent distillation stage.

3.9 Raw-CDF distillation and onset refinement

The NLL stage produces well-calibrated heteroscedastic predictions on the fitted training targets, but the fitted curves are not equally faithful over the whole time axis. Near the onset of injection the absolute penetration is small, and any timing offset, fit-delay ambiguity, or smoothness prior produces a relatively large fractional error, so the earliest segment of each predicted curve is dominated by the fitted-target prior rather than by directly observed raw plume dynamics. Figure 12 illustrates this in a representative toy-inference sweep.

To refine the early-time dynamics against raw CDF observations while preventing collapse in the late-time sparse regime, a knowledge-distillation stage is implemented in MLP/distillation_plus_raw_series.py and archived under runs_mlp/distill_cdf_onset_v2_20260410_103413. The Stage-2 NLL model serves as a fixed teacher; the student network inherits the teacher weights and is further trained on raw CDF together with teacher guidance.

CDF coverage regimes. Because the raw CDF series are right-censored (by $t \approx 3$ ms many operating conditions have lost more than 70% of their measurements), direct raw supervision cannot be applied uniformly in time. For each operating-condition group, raw observations are binned in $\Delta t = 0.1$ ms intervals over $[0, 5]$ ms, and the smoothed sample count per bin determines three mutually exclusive regimes: `raw_reliable` (smoothed coverage $\geq 70\%$ of peak), `raw_uncertain` (coverage



images/stage2_toy_inference_2000bar_5bar.png

Figure 12: Toy inference of the stage-2 NLL model for a representative 2000 bar, 5 bar case. The prediction is smooth and well-behaved overall, but the earliest part of the trajectory is still shaped primarily by the fitted-target prior, motivating the subsequent distillation refinement.

between 20% and 70%), and `teacher_only` (coverage below 20% or fewer than 4 samples per bin). Transitions require at least two consecutive bins in a regime to prevent isolated dips from triggering premature regime changes.

Supervision weights. Each grid point receives two independent weight channels:

Regime	Raw weight w_{raw}	KD weight w_{kd}
<code>raw_reliable</code>	1.0	0.25
<code>raw_uncertain</code>	0.0	1.0
<code>teacher_only</code>	0.0	0.75

Both channels are further modulated by a global time-bin rebalancing weight (inverse-frequency, clipped to $[0.5, 2.0]$). In `raw_reliable` bins, raw NLL is the primary signal with mild KD regularisation. In `raw_uncertain` bins, raw supervision is dropped and teacher KL dominates. In `teacher_only` bins, the KD weight is slightly reduced to acknowledge teacher extrapolation uncertainty.

Composite loss. The training objective is

$$\mathcal{L} = \mathcal{L}_{\text{raw}} + \mathcal{L}_{\text{KD}} + \lambda_{\text{onset}} \mathcal{L}_{\text{onset}} + \lambda_{\text{anchor}} \mathcal{L}_{\text{anchor}} + \lambda_{d_1} P_{d_1} + \lambda_{d_2} P_{d_2},$$

where $\mathcal{L}_{\text{raw}}$ is the Gaussian NLL on physical (μ, σ^2) against raw CDF targets weighted by w_{raw} ; $\mathcal{L}_{\text{KD}}$ is the forward KL divergence $\text{KL}(q_{\text{student}} \| p_{\text{teacher}})$ in physical space weighted by w_{kd} and a teacher confidence factor

$$w_{\text{conf}}(t) = \text{clamp}\left(\frac{\sigma_{\text{ref}}}{\sigma_{\text{teacher}}(t)}, 0.25, 1.0\right), \quad \sigma_{\text{ref}} = 10 \text{ mm};$$

$\mathcal{L}_{\text{onset}}$ is a BCE loss on a new onset logit output against a ramp target $\min(t/t_{\text{ramp}}, 1)$ active only for $t \leq 0.2 \text{ ms}$ ($\lambda_{\text{onset}} = 0.1$); $\mathcal{L}_{\text{anchor}}$ is a quadratic penalty pulling $\mu(t \approx 0)$ toward zero, ramping down over the first 0.15 ms ($\lambda_{\text{anchor}} = 0.01$); $P_{d_1} = \text{ReLU}(-\Delta\mu)^2$ enforces monotonicity; and $P_{d_2} = \text{ReLU}(\Delta^2\mu)^2$ enforces concavity for $t \geq 0.5 \text{ ms}$.

Student architecture. The student is a three-output MLP extending the teacher’s two outputs $[\hat{\mu}, \log \hat{\sigma}^2]$ with an onset logit. All shared layers are copied from the teacher checkpoint; the final linear row for the onset head is initialised to zero (sigmoid $\rightarrow 0.5$). All losses are computed in physical space by multiplying student and teacher outputs by the per-sample A before evaluating $\mathcal{L}_{\text{KD}}$, $\mathcal{L}_{\text{raw}}$, and the shape priors.

Training and results. The student is optimised with AdamW at $\text{lr} = 3 \times 10^{-3}$ with gradient clipping at norm 1.0 and early stopping with patience 20. In the archived run `distill_cdf_onset_v2_20260410_103413`, the best validation loss was 2.327 at epoch 55 (val raw NLL 2.256, val KD KL 0.018, val onset BCE 0.308). On the held-out test set: raw NLL 2.29, KD KL 0.019, onset BCE 0.309, with negligible derivative penalties ($P_{d_1} \approx 6.5 \times 10^{-7}$, $P_{d_2} \approx 2.3 \times 10^{-5}$). The run directory contains `best_model_refinement.pt` and `loss_curves.png`; Figure 13 shows the training curves.

3.10 Prototype inference notebook

The notebook `MLP/toy_inference_from_run.ipynb` demonstrates how a stored run directory can be loaded for manual inference. The archived output shows that, when pointed to the stage-2 run directory above, the prototype resolves the available checkpoint as `best_model_stage1.pt`. It then generates mean and standard-deviation curves and overlays them on simple geometric masks.

This prototype is useful as a proof of concept because it exercises the scaling state, the trained network interface, and a simple downstream visualization path. It is not used here as evidence for a finished collision-detection product.

4 Results

4.1 Extracted and cleaned trajectory assets

The current workspace contains ten campaign folders under `MLP/synthetic_data` and 295 cleaned fitted CSV files under the corresponding `clean` subfolders. This



Figure 13: Distillation stage training curves from `runs_mlp/distill_cdf_ons_et_v2_20260410_103413/loss_curves.png`. The composite loss converges to a best validation of 2.327 at epoch 55.

means the image-processing side of the workflow is not merely a draft: it has already produced a sizeable archive of cleaned penetration-oriented assets from multiple campaign folders.

Figure 14 shows a representative cleaned penetration plot from the archived output directory. The solid curves are cleaned trajectories and the dashed curves are the fitted reduced-order curves from the `cdf` pathway. The figure illustrates the practical role of `MLP/fit_raw_data.py`: smooth the raw plume-wise trajectories enough that they can be modeled later without discarding their main temporal structure.

4.2 Stage-1 MSE baseline

The Stage-1 MSE baseline establishes the representative-trajectory mean model. Table 3 summarises the quantitative evidence across all three training stages.

The archived run `stage1_engineered_mse_a_only_20260410_020317`



Figure 14: Representative cleaned trajectories and fitted curves from MLP/synthetic_data/BC20241003_HZ_Nozzle1/cdf/plots_clean/T1.png. The figure is used here as direct evidence that the trajectory-cleaning and fitting stage has already produced reusable penetration targets.

trained on 526 representative trajectories, selected as the group-median curve at $t = 5$ ms after fit-level masking. The run achieved a best validation MSE of 18.94 on the scaled target at epoch 115, with a test physical MAE of 8.1 mm. The pretrain collapse report confirms a 59.5% post-0.5 ms variance reduction under the A -scaling hypothesis, validating that the scaling accounts for the dominant diameter and pressure variation before the network sees any data. The run directory contains `best_model_stage1.pt`, `scaler_state.json`, epoch and iteration loss logs, and `pretrain_collapse_summary.png`.

4.3 Stage-2 NLL extension

The Stage-2 run expands the training set to all 14,594 filtered trajectories and replaces the MSE objective with heteroscedastic NLL, warm-starting from the Stage-1



Figure 15: Stage-1 training curves from MLP/runs_mlp/stage1_engineered_mse_a_only_20260410_020317/stage1_loss_curves.png.

checkpoint.

The archived run `stage2_engineered_nll_a_only_20260410_020607` reached a best validation NLL of 2.038 at epoch 66. On the held-out test set the physical MAE is 9.2 mm and the mean predicted standard deviation is 11.4 mm, indicating that the model assigns nontrivial uncertainty broader than the MAE — consistent with genuine conditional variability in the filtered dataset. The run directory contains `best_model_stage2.pt` and serves as the teacher checkpoint for the distillation stage.

4.4 Distillation refinement

The distillation stage archives a student that has been refined toward raw CDF observations in the well-covered time region while remaining anchored to the teacher where coverage is sparse.

The archived run `distill_cdf_onset_v2_20260410_103413` reached a best

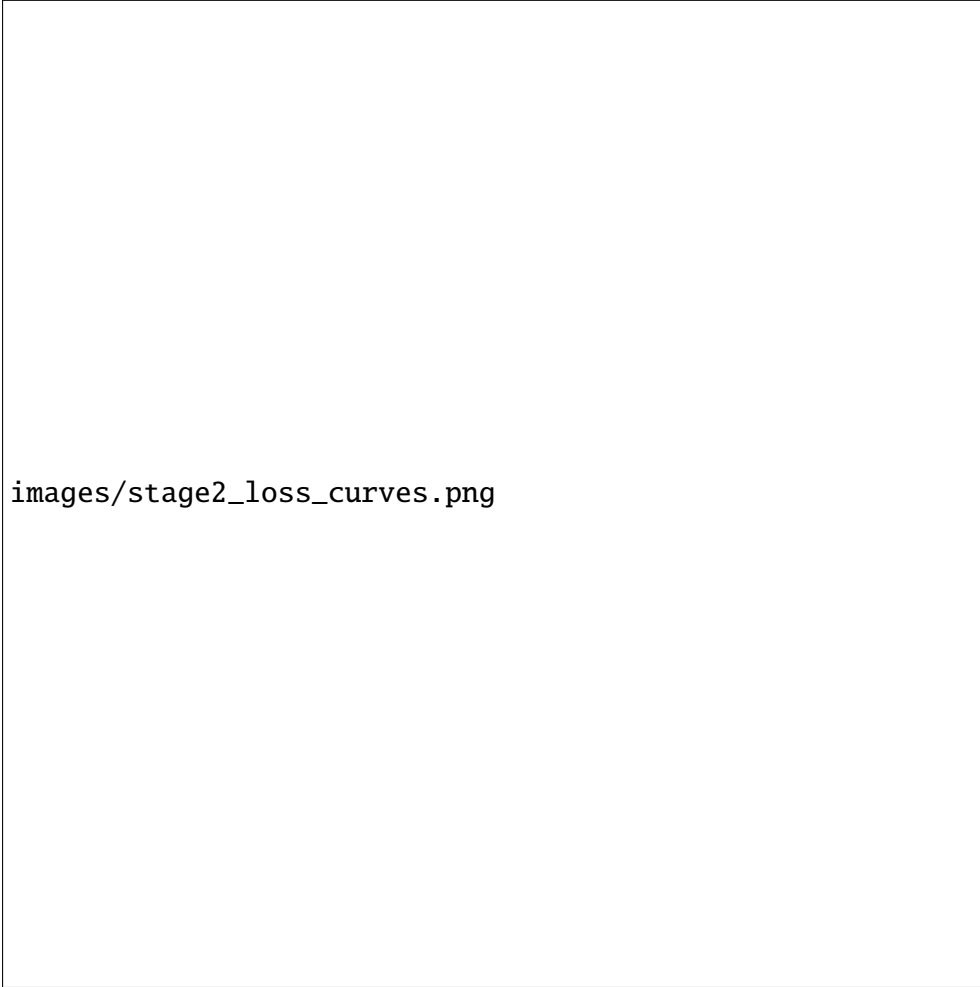


Figure 16: Stage-2 training curves from `MLP/runs_mlp/stage2_engineered_nll_a_only_20260410_020607/stage2_loss_curves.png`.

validation composite loss of 2.327 at epoch 55 (val raw NLL 2.256, val KD KL 0.018, val onset BCE 0.308). On the held-out test set: raw NLL 2.29, KD KL 0.019, onset BCE 0.309, with negligible derivative penalties ($P_{d_1} \approx 6.5 \times 10^{-7}$, $P_{d_2} \approx 2.3 \times 10^{-5}$). The run directory contains `best_model_refinement.pt`, `epoch_loss.csv`, and `loss_curves.png`; the training curves are shown in Figure 13.

4.5 Prototype inference notebook

The prototype inference notebook demonstrates a plausible deployment direction rather than a final scientific endpoint. Its archived outputs show that it loads run metadata, produces predicted mean and standard-deviation curves, draws simple spray-envelope proxies, and computes simple overlap metrics with wall and cylinder masks. In the stored example the notebook reports zero wall-collision and cylinder-collision area for the chosen toy setup. That is useful as a software integration proof of concept, but it is not enough to claim a validated collision-analysis framework.

Stage	Data evidence	Best validation result	Current status
Image-processing and fitting	10 campaign folders and 295 cleaned fitted CSV files under <code>MLP/synthetic_data</code>	Representative cleaned/fitted trajectory plots available	Completed and auditable
Stage 1 MSE baseline	526 representative rows, group-aware split	Val MSE 18.94 at epoch 115; test MAE 8.1 mm	Completed with <code>best_model_stage1.pt</code>
Stage 2 NLL extension	14,594 filtered rows, warm-started from Stage 1	Val NLL 2.038 at epoch 66; test MAE 9.2 mm	Completed with <code>best_model_stage2.pt</code>
Distillation refinement	Raw CDF + teacher KD, regime-dependent supervision	Val composite 2.327 at epoch 55	Completed with <code>best_model_refinement.pt</code>
Prototype inference	Archived notebook outputs from <code>MLP/toy_inference_from_run.ipynb</code>	Mean/std prediction plots and simple overlap calculations	Prototype only, not a validated product

Table 3: Evidence summary of the completed project stages.

4.6 Result boundary and reproducibility status

The evidence in this thesis comes from three levels:

1. directly present assets in the current workspace, such as cleaned CSV files, archived figures, and run logs;
2. archived notebook outputs embedded inside the stored notebooks; and
3. stored run configurations and checkpoints.

All three training stages have dedicated deployable checkpoints in the workspace: Stage-1 `best_model_stage1.pt`, Stage-2 `best_model_stage2.pt`, and distillation `best_model_refinement.pt`, together with their scaler states, training

configurations, and epoch-level loss logs. The extraction, cleaning, and three-stage surrogate pipeline are fully archived and auditable. The prototype inference notebook has not been upgraded to a validated product and remains the primary open boundary.

5 Conclusions

This thesis documented an evidence-grounded workflow for processing high-speed Mie-scattering spray videos into penetration-oriented engineering observables and for building a complete three-stage surrogate model on top of those observables. The completed contributions are:

- manual nozzle-center calibration in `GUI.py`,
- batch plume-wise feature extraction in `main.py`,
- trajectory cleaning and reduced-order fitting in `MLP/fit_raw_data.py`,
- a v2 engineered-feature architecture that replaces sparse pressure and diameter inputs with the Hiroyasu–Arai scaling group $A = (\Delta P / \rho_a)^{1/4} d_n^{1/2}$, validated by a pretrain collapse check showing 59.5% post-0.5 ms variance reduction,
- a completed Stage-1 deterministic MSE baseline (val MSE 18.94 at epoch 115, test MAE 8.1 mm) with group-aware splits and a saved checkpoint,
- a completed Stage-2 heteroscedastic NLL extension (val NLL 2.038 at epoch 66, test MAE 9.2 mm, mean std 11.4 mm) with a dedicated checkpoint serving as the distillation teacher, and
- a completed knowledge-distillation refinement stage (val composite loss 2.327 at epoch 55) that aligns early-time predictions toward raw CDF observations using regime-dependent supervision while preserving late-time stability through teacher KL anchoring.

The three-stage surrogate pipeline is complete as an archived engineering artifact. The physical A -scaling assumption was empirically validated by the pretrain collapse check, and the regime-dependent distillation framework provides a principled way to incorporate right-censored raw CDF data without biasing the model toward disappearing late-time observations. What remains open is the transition from archived prototype to production deployment: validated collision prediction against explicit experimental or simulation targets, full 2D envelope reconstruction, and multi-run reproducibility studies are the natural next development priorities.